

## PROFESSIONAL EXPERIENCE

- Automation Engineering Intern — Lam Research — Bangalore, India** Jun 2025 – Dec 2025
- Architected 15 enterprise automation solutions using Python, SAP ABAP, JavaScript, and C# across Proto4lab, Pilot MPM, CPE, GSCM, and CEM departments, eliminating 2000+ hours of manual work annually
  - Engineered ZERF Data Automation System integrating SAP GUI Scripting with Python win32com, transforming 3-hour daily workflows into 15-minute streamlined processes while processing 500+ transactions at 99.9% accuracy
  - Built Power BI analytics dashboard processing 10,000+ weekly data points for Project Gold Mine, improving NPI Supply Chain visibility by 70% through real-time SAP data extraction and automated reporting
  - Created Chrome extension using JavaScript for automatic BOM extraction and engineering drawing management, reducing download time by 60% and standardizing workflows across 50+ engineers
- UAV Engineering Intern — NDRF 8th Battalion — Uttar Pradesh, India** Jan 2025 – Mar 2025
- Designed "Rakshak" autonomous drone achieving 80% obstacle avoidance accuracy using PixHawk and Raspberry Pi
  - Deployed ADAS modules for real-time aerial assessment supporting 30+ million disaster-affected population
- Backend & AI Engineering Intern — National Informatics Center — Delhi, India** Jul 2024 – Sep 2024
- Constructed AI authentication system using YOLOv8 and Vision Transformer for National Awards Portal (300,000+ users)
  - Achieved 89% model accuracy through iterative A/B testing and production deployment optimization

## RESEARCH PROJECTS

- LostInEffect - Quantum-Resistant Cryptography — Python, TensorFlow, Delithium** GitHub/private repo
- Pioneering novel quantum-resistant technology leveraging chaos theory and butterfly effect algorithms, achieving computational irreversibility against quantum threats (research paper in progress)
- BEYOND GENERATION - Hyperspectral Anomaly Detection — PyTorch, LRSR, EBT** GitHub/private repo
- Introduced first-ever integration of Energy-Based Transformers with Low-Rank Sparse Representation for AI Grand Challenge, processing 30km satellite imagery with top-tier PR-AUC scores (research paper in progress)

## INDUSTRIAL PROJECTS

- CatalogCraft - National Champion, Build for Bharat — Kotlin, React, Django, PostgreSQL** GitHub
- Won National Championship at Google-Antler-Paytm-ONDC hackathon by digitizing 1000+ SKUs in 17 languages, improving search accuracy by 35% through OCR, voice recognition, and YOLO object detection
- 4-DOF Robotic System - Lam Research Challenge Finalist — C++, ROS, ESP32** GitHub
- Selected among top 25 teams nationally for Lam Research Challenge 2024, designed precision temperature control achieving  $\pm 1.0^{\circ}\text{C}$  accuracy with PID-controlled Peltier cooling (project led directly to internship offer)
- Amazon ML Challenge 2024 — XGBoost, BLIP, CLIP, Transformers** GitHub
- Benchmarked 4 state-of-the-art models achieving 90% multimodal accuracy in 72-hour intensive learning sprint

## TECHNICAL SKILLS

- Languages:** Python, JavaScript, Java, C++, C#, TypeScript, SQL, Rust, Kotlin, Solidity, R, PHP
- Frameworks:** React, Django, Node.js, FastAPI, Flask, Express.js, Next.js, Flutter, Laravel, TensorFlow, PyTorch
- Cloud/DevOps:** AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Vercel, Firebase
- Data/AI:** pandas, NumPy, scikit-learn, OpenCV, Keras, Power BI, Tableau, SAP ABAP, Selenium
- Databases:** PostgreSQL, MongoDB, MySQL, Redis, Supabase, SQLite, MariaDB

## EDUCATION

- Bachelor of Technology in Computer Science** Expected May 2026  
Maharaja Agrasen Institute of Technology, New Delhi
- Kendriya Vidyalaya R.K. Puram, New Delhi — Mathematics Olympiad (2x), Science Olympiad Winner** 2010 – 2022

## LEADERSHIP & ACHIEVEMENTS

- Led teams to finals in 60+ hackathons as Technical Lead, demonstrating consistent innovation and strategic execution
- Founded BYTE Tech Society (100+ members), organizing 10+ workshops and mentoring 200+ students in emerging tech
- BYTE Tech Society President, T-Hub National Hackathon Finalist, Lam Research Challenge 2024 Finalist, Open-source contributor